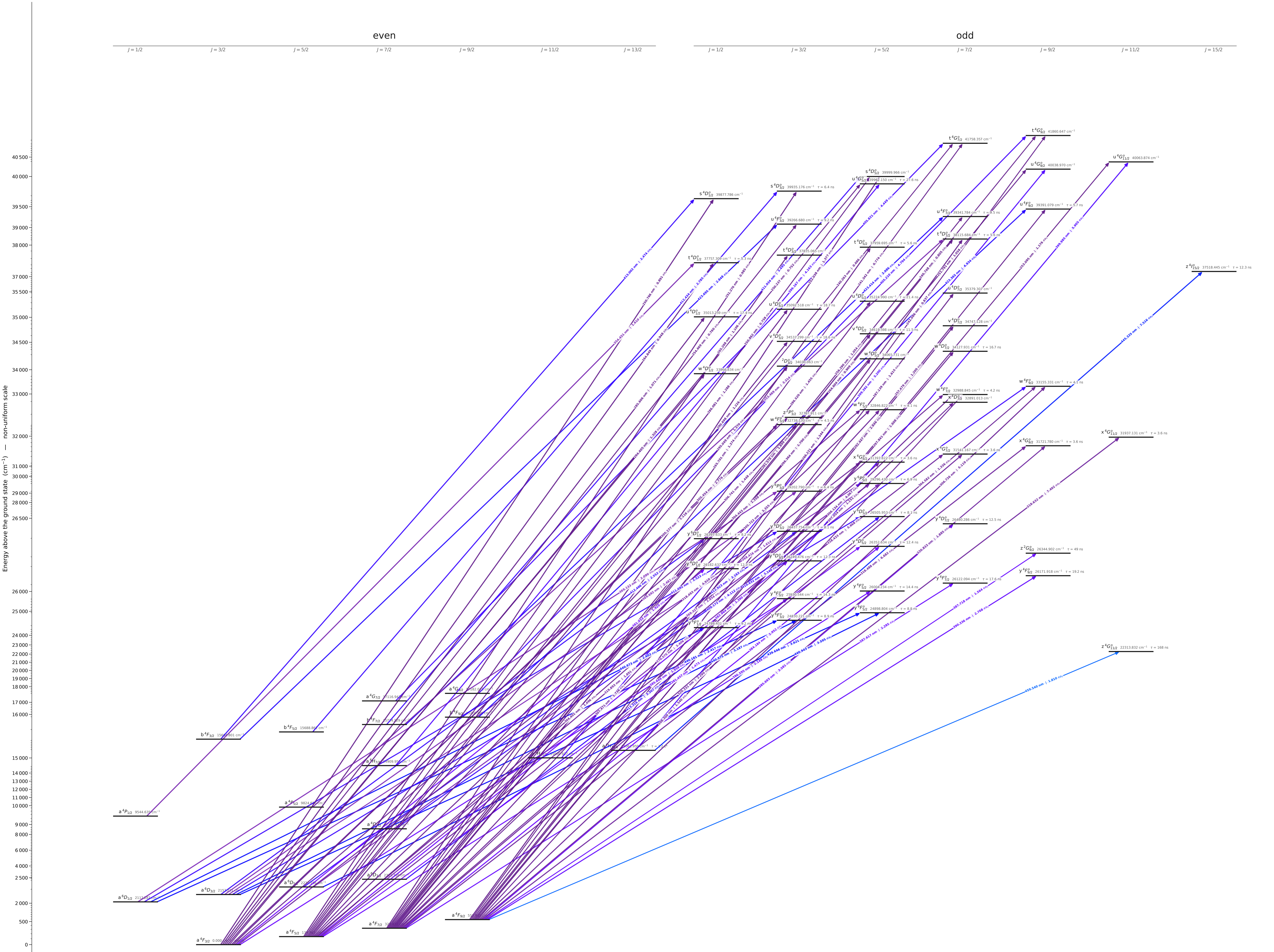


V⁺ ionisation limit 54411.67 cm⁻¹ = 6.74619 eV (183.784 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle f|er|f'\rangle|$ in $e a_0$
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
 Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists)



Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ U e J J $ (ea0)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$a^4H_{132} - z^4F_{15Q}$	445.3255	445.2005	673.1985	7.516	8.100e+07	12.94	100	laser-cooling

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)									
$y^3P_{1/2}$	9930		F=1: -112.98	F=2: -915.12	F=3: -618.32	F=4: -222.60	F=5: +272.06	F=6: +865.65				
$y^3D_{5/2}^o$	1530		F=1: -168.63	F=2: -138.65	F=3: -93.69	F=4: -33.73	F=5: +41.22	F=6: +131.16				
$y^3P_{3/2}$	54060		F=2: -3642.48	F=3: -2023.60	F=4: +134.91	F=5: +2833.04						
$y^3P_{1/2}^o$	39060		F=1: -4381.48	F=2: -3505.09	F=3: -2465.81	F=4: -876.89	F=5: +1071.76	F=6: +3410.14				
$x^3G_{5/2}$	60310		F=1: -6779.06	F=2: -5573.89	F=3: -3743.14	F=4: -1355.81	F=5: +1657.10	F=6: +5272.60				
$x^3F_{7/2}^o$	14600		F=2: -4580.08	F=3: -3707.69	F=4: -2519.29	F=5: -1677.09	F=6: -167.09	F=7: +1526.69	F=7: +3562.69			
$x^3F_{5/2}^o$	15900		F=1: -3058.63	F=2: -2740.65	F=3: -264.18	F=4: -1628.62	F=5: -83.14	F=6: -119.17	F=7: +1231.40	F=8: +2502.52		
$x^3F_{3/2}^o$	81100		F=2: -1841.48	F=3: -1598.64	F=4: -1274.87	F=5: -870.75	F=6: -384.48	F=7: +182.12	F=8: +829.68	F=9: +1558.17		
$w^3F_{5/2}$	25800		F=1: -2900.50	F=2: -2384.85	F=3: -1611.34	F=4: -580.10	F=5: +709.01	F=6: +2255.94				
$w^3F_{3/2}$	16900		F=2: -1699.82	F=3: -1591.89	F=4: -1376.04	F=5: -1052.27	F=6: -620.57	F=7: -620.57	F=8: +566.61	F=9: +1322.08		
$z^3F_{7/2}^o$	18130		F=4: -5440.47	F=5: -4526.11	F=6: -3428.87	F=7: -2148.76	F=8: -685.77	F=9: +960.08	F=10: +2788.81	F=11: +4800.42		
$z^3F_{5/2}^o$	21300		F=2: -1436.76	F=3: -798.20	F=4: +53.21	F=5: +1117.48						
$t^3D_{5/2}^o$	14400		F=0: -2266.43	F=1: -2122.53	F=2: -1834.72	F=3: -1403.03	F=4: -827.42	F=5: -107.92	F=6: +755.48	F=7: +1467.85		
$u^3F_{5/2}$	28530		F=2: -5482.46	F=3: -4912.85	F=4: -3405.84	F=5: -2491.23	F=6: -1495.22	F=7: +213.60	F=8: +2207.12	F=9: +7742.78		

Measured A, B constants; sources in the data file